CHOW'S LEMMA FOR FINITE TYPE MORPHISMS OVER LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. No omissions in the counted span: the reduction is worked; the Chow core is a precise citation of a proved lemma.

Allowed standard prerequisites (editorial).

- Noetherian Chow lemma. Cohomology of Schemes, Lemma coherent-lemma-chow-Noetherian
- Finite type closed in fp. Limits, Lemma lemma-finite-type-closed-in-finite-presentation (same chapter)

Source reference. Original proposition 01ZJ: sources/stacks-chow/tags.txt, source lines 2596–2596. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

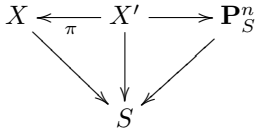
Source reference. Original proposition 01ZA: sources/stacks-chow/tags.txt, source lines 2585–2585. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZQ: sources/stacks-chow/tags.txt, source lines 2603–2603. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0200: sources/stacks-chow/tags.txt, source lines 2617–2617. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a quasi-compact and quasi-separated scheme. Let $f : X \to S$ be a separated morphism of finite type. Then there exists an $n \geq 0$ and a diagram*



where $X' \to \mathbf{P}^n_S$ is an immersion, and $\pi : X' \to X$ is proper and surjective.

Proof. By Proposition 01ZJ we can find a closed immersion $X \to Y$ where Y is separated and of finite presentation over S. Clearly, if we prove the assertion for Y, then the result follows for X. Hence we may assume that X is of finite presentation over S.

Write $S = \lim_i S_i$ as a directed limit of Noetherian schemes, see Proposition 01ZA. By Lemma 01ZM we can find an index $i \in I$ and a scheme $X_i \to S_i$ of finite presentation so that $X = S \times_{S_i} X_i$. By Lemma 01ZQ we may assume that $X_i \to S_i$ is separated. Clearly, if we prove the assertion for X_i over S_i, then the assertion

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holds for X. The case $X_i \to S_i$ is treated by Cohomology of Schemes, Lemma 0200. $\square$